SABARI M

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EXECUTIVE SUMMARY

Electronics and Communication Engineering student with hands-on experience in PCB design and exploring analog circuits. Proficient in Altium Designer and STM32 development tools.

SKILLS

PCB Design: Altium Designer, Schematic capture, Circuit Design, PCB layout Design, DRC, Gerber generation, Output job files

Embedded Systems: STM32, ESP32, Sensor calibration

Programming Language: C, MATLAB

Tools & Tech: STM32CubeIDE, Altium Designer, KiCAD, Arduino IDE, STM32CubeMonitor, Keil uVision, VS Code, ANSYS HFSS, Verilator.

INTERNSHIP EXPERIENCE

Punha LLP, *PCB Design intern* — 03/2026 – 06/2026 | Remote

- Working in designing circuits and PCB for healthcare wearables
- Explored ADC ICs for designing PCBs for prosthetic wearable devices

ISRO - LIQUID PROPULSION SYSTEM CENTER, *Student intern* — 10/2025 – 11/2025 | Tiruvandrum, Kerala
Intern – Flow Measurement & Sensor Calibration

- Worked on calibration of **Coriolis** and **Thermal Mass Flow Meters sensors** using standard **instrumentation** practices.
- Studied flow measurement chains including **sensors**, **transmitters**, **signal conditioning**, **TIB**, and **DAQ systems**.
- Gained hands-on exposure to **analog signal processing**, **sensor interfacing**, and **industrial instrumentation electronics**.

MINI PROJECTS

CUSTOM ESP32S3 MINI WITH USB TYPE C - PCB DESIGN

- Designed a custom ESP32-S3 Mini development board integrating USB Type-C for efficient power and data transfer.
- Optimized the PCB for compact form factor and manufacturability, targeting IoT, embedded systems, and wireless applications.

Orientation Control Using MPU6050 by STM32F401CCU6 MCU

- Acquired raw accelerometer and gyroscope data, applied offset calibration, and calculated roll angle from accelerometer readings.
- Implemented a Kalman filter in C to fuse accelerometer and gyroscope data, achieving smooth and precise tilt angle estimation.

Automatic Temperature and Water Controlling System

- Developed a microcontroller-based system to monitor and control temperature and water levels using appropriate sensors.
- Implemented sensor interfacing and control logic on Arduino for real-time measurement and automated actuation.
- Simulated and validated the system design using TinkerCAD before hardware implementation.

CERTIFICATES

Altium Skilled Engineer

Altium Emerging Engineer

Altium Designer Essential - On demand

Altium Workspace User Management

PCB Design Essentials - Altium

C Programming Hands On : Skill Rack

Embedded Systems

EDUCATION

Rajalakshmi Institute of Technology, Chennai, 2023 – 2027
B.E , Electronics and Communication Engineering

Sir .C. P. Memorial Government Higher Secondary School 2016 – 2023
Kanyakumari, VI to XII
HSC Percentage -89%

ORGANISATIONS

Rajalakshmi Institute of Technology, 09/2025 – Present
IEEE Student Branch Treasurer

IEEE Antennas and Propagation Society, *Student Member* 06/2025 – Present